

Total No. of Questions : 4]

SEAT No. :

PE-198

[Total No. of Pages : 3

[6580]-558

B.E. (Mechanical) (Insem.)
HEATING VENTILATION AIR CONDITIONING &
REFRIGERATION
(2019 Pattern) (Semester - VII) (402041)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Answer Q1 or Q2, Q3 or Q4.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Use of electronic pocket calculator is allowed.
- 5) Use of steam table and psychrometric chart is allowed.

Q1) a) Explain the concept of DART in air refrigeration cycles and compare various air refrigeration cycles using DART. **[6]**

b) The following data refers to a bootstrap air cycle evaporative refrigeration cycle used for an evaporator to take 20 tonnes of refrigeration load : **[9]**

Ambient air temperature = 15°C

Ambient air pressure = 0.8 bar

Mach number of flight = 1.2

Ram efficiency = 90%

Pressure of air after main compressor = 4 bar

Pressure of air after secondary compressor = 5 bar

Isentropic efficiency of main compressor = 90%

Isentropic efficiency of secondary compressor = 80%

Isentropic efficiency of cooling turbine = 80%

Temperature of air leaving the first heat exchanger = 170°C

Temperature of air leaving the second heat exchanger = 155°C

Cabin pressure = 1 bar :

Cabin temperature = 25°C

Find :

- i) Draw neat temperature entropy plot of given system
- ii) Mass of air required to take cabin load
- iii) Power required for the refrigeration system
- iv) C.O.P. of the system

P.T.O.

OR

Q2) a) Explain the following properties of refrigerants [8]

- i) Latent heat of vaporization
- ii) Boiling point
- iii) Miscibility
- iv) Specific heat of vapour refrigerant

b) Explain Boot -Strap air refrigeration system with T-S diagram [7]

Q3) a) Explain two stage compression with Intercooling and sub cooling by external cooling source. [6]

b) A Single Compressor using R-12 as Refrigerant as three evaporators of capacity 10 TR, 20 TR, 30 TR. All evaporators operate at -10°C and the vapors within the compressor are dry and saturated. The condenser temperature is 40°C , the liquid refrigerant leaving the condenser is sub cooled to 30°C . [9]

Assuming the Isentropic Compression, Find

- i) Mass flow rate of refrigerant flowing through each evaporator.
- ii) The power required to drive the compressor
- iii) COP of the system

OR

Q4) a) Explain with schematic and P-h diagram a complete two stage compression system with flash chamber acts as intercooler and gas removal. [6]

b) With the neat diagram Explain working of Cascade refrigeration system. [9]

